

Ex.No. 12: NB-IoT vs LTE-M Throughput Simulation**Date:1-8-26**

AIM: To simulate and compare the performance of NB-IoT and LTE-M communication technologies in terms of Throughput (Mbps), Latency (ms), and Packet Delivery Ratio (%) using MATLAB.

Objective:

1. To evaluate and compare the Throughput (Mbps) performance of NB-IoT and LTE-M under varying network conditions.

2.

To analyze the Latency (ms) characteristics of NB-IoT and LTE-M and study their impact on real-time communication.

3.

To measure and compare the Packet Delivery Ratio (%) of NB-IoT and LTE-M systems for reliability assessment.

Procedure

1. Define system parameters for NB-IoT and LTE-M such as bandwidth, transmission power, and modulation efficiency.

2. Generate network conditions by varying Signal-to-Noise Ratio (SNR) or channel quality values.

3. Compute Throughput (Mbps) for both NB-IoT and LTE-M using their respective data rate models.

4. Calculate Latency (ms) based on transmission time, scheduling delay, and network load.

5. Determine Packet Delivery Ratio (PDR %) by comparing successfully received packets with transmitted packets.

6. Simulate multiple iterations to observe performance under different channel conditions.

7. Plot comparison graphs for Throughput, Latency, and PDR between NB-IoT and LTE-M.

8. Analyze results to identify which technology performs better under different IoT scenarios.

Objective 1 Matlab Code and Output:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Throughput (Mbps)
NB_IoT = 0.05*(1-exp(-SNR/8));
LTE_M = 1.2*(1-exp(-SNR/8));

disp('SNR (dB) NB-IoT(Mbps) LTE-M(Mbps)')
disp([SNR' NB_IoT' LTE_M'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Throughput vs SNR')
```

```

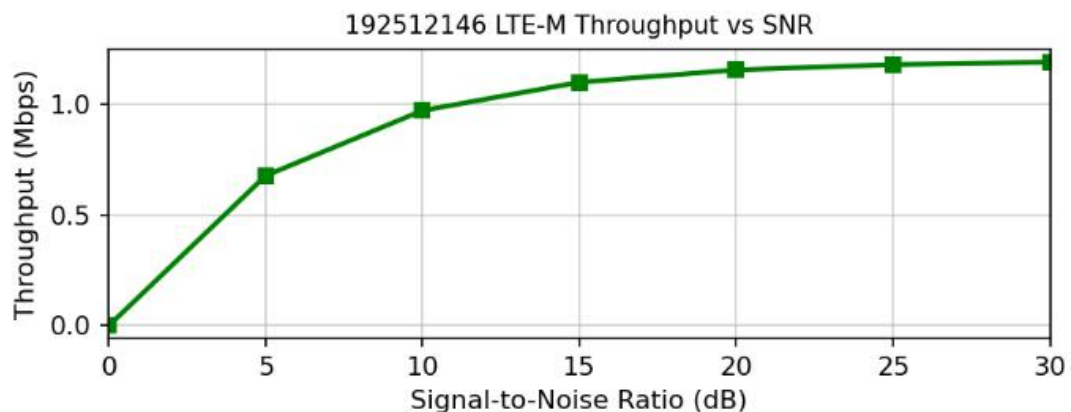
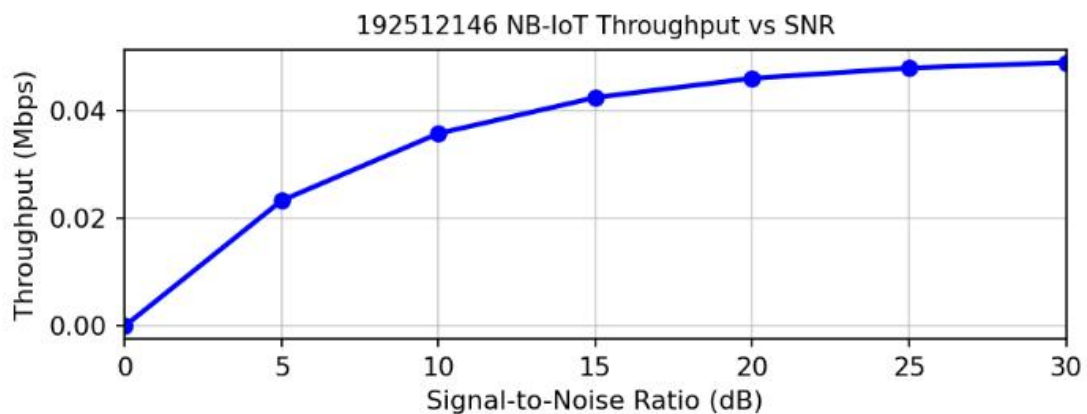
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on

```

```

exp121.m
MATLAB Drive/exp121.m
1  clc;
2  clear;
3  close all;
4  % SNR values (dB)
5  SNR = 0:5:30;
6  % Simulated Throughput (Mbps)
7  NB_IoT = 0.05*(1-exp(-SNR/8));
8  LTE_M = 1.2*(1-exp(-SNR/6));
9
10 disp('SNR (dB)  NB-IoT(Mbps)  LTE-M(Mbps)')
11 disp([SNR' NB_IoT' LTE_M'])
12 figure
13 % ----- Graph 1 -----
14 subplot(2,1,1)
15 plot(SNR,NB_IoT,'-o','LineWidth',2,'Color','b')
16 title('192512146 NB-IoT Throughput vs SNR')
17 xlabel('Signal-to-Noise Ratio (dB)')
18 ylabel('Throughput (Mbps)')
19 grid on
20 % ----- Graph 2 -----
21 subplot(2,1,2)
22 plot(SNR,LTE_M,'-s','LineWidth',2,'Color','g')
23 title('192512146 LTE-M Throughput vs SNR')
24 xlabel('Signal-to-Noise Ratio (dB)')
25 ylabel('Throughput (Mbps)')
26 grid on

```

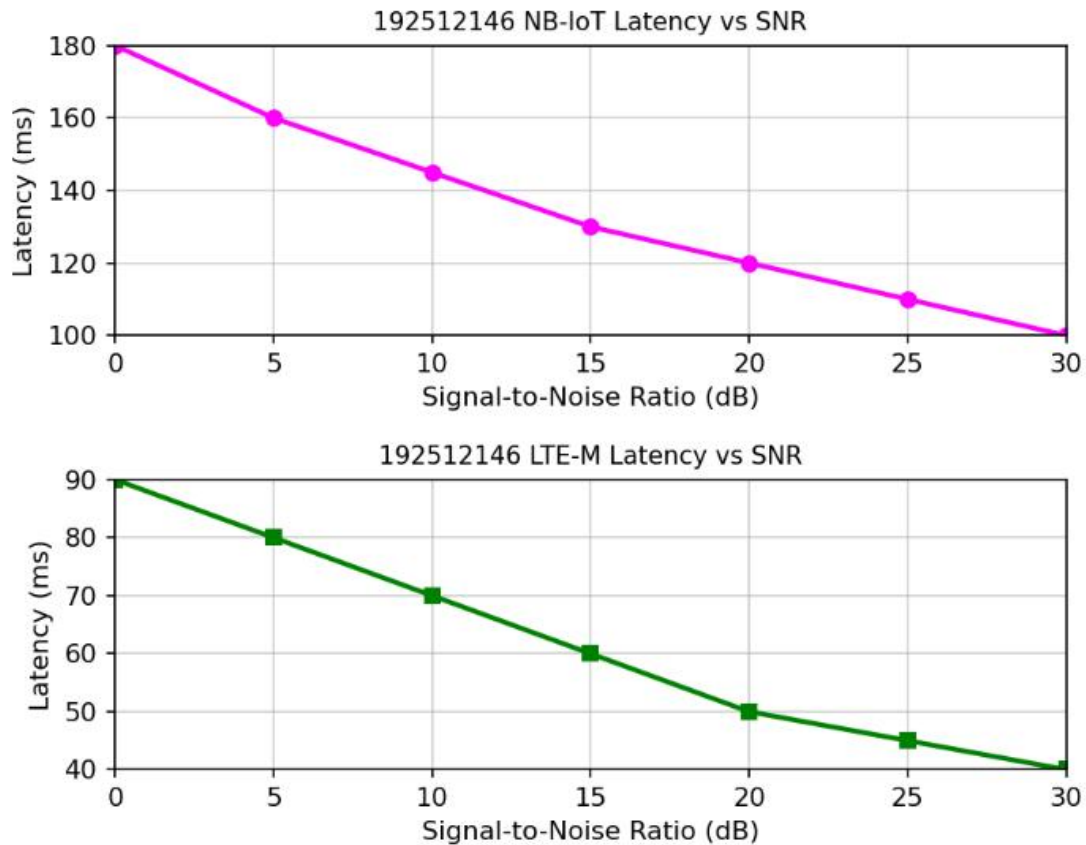


The throughput of both NB-IoT and LTE-M increases as the Signal-to-Noise Ratio (SNR) improves. LTE-M achieves significantly higher throughput than NB-IoT, making it more suitable for applications requiring higher data rates.

Objective 2 Matlab Code and Output:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Latency (ms)
NB_IoT = [180 160 145 130 120 110 100];
LTE_M = [90 80 70 60 50 45 40];
disp('SNR(dB) NB-IoT Latency(ms) LTE-M Latency(ms)')
disp([SNR' NB_IoT' LTE_M'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on
```

```
exp122.m
MATLAB Drive/exp122.m
1  clc;
2  clear;
3  close all;
4  % SNR values (dB)
5  SNR = 0:5:30;
6  % Simulated Latency (ms)
7  NB_IoT = [180 160 145 130 120 110 100];
8  LTE_M = [90 80 70 60 50 45 40];
9  disp('SNR(dB) NB-IoT Latency(ms) LTE-M Latency(ms)')
10 disp([SNR' NB_IoT' LTE_M'])
11 figure
12 % ----- Graph 1 -----
13 subplot(2,1,1)
14 plot(SNR,NB_IoT,'-o','LineWidth',2,'Color','magenta')
15 title('192512146 NB-IoT Latency vs SNR')
16 xlabel('Signal-to-Noise Ratio (dB)')
17 ylabel('Latency (ms)')
18 grid on
19 % ----- Graph 2 -----
20 subplot(2,1,2)
21 plot(SNR,LTE_M,'-s','LineWidth',2,'Color','g')
22 title('192512146 LTE-M Latency vs SNR')
23 xlabel('Signal-to-Noise Ratio (dB)')
24 ylabel('Latency (ms)')
25 grid on
```



The latency of both NB-IoT and LTE-M decreases as the SNR increases due to improved channel quality. LTE-M exhibits lower latency than NB-IoT, making it more suitable for delay-sensitive IoT applications such as real-time monitoring and asset tracking.

Objective 3 Matlab Code and Output:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Packet Delivery Ratio (%)
NB_IoT = [82 86 89 92 95 97 99];
LTE_M = [88 91 94 96 98 99 100];
disp('SNR(dB) NB-IoT PDR(%) LTE-M PDR(%)\n')
disp([SNR' NB_IoT' LTE_M'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
```

% ----- Graph 2 -----

```
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
```

```
exp123.m
MATLAB Drive/exp123.m
1  clc;
2  clear;
3  close all;
4  % SNR values (dB)
5  SNR = 0:5:30;
6  % Simulated Packet Delivery Ratio (%)
7  NB_IoT = [82 86 89 92 95 97 99];
8  LTE_M = [88 91 94 96 98 99 100];
9  disp('SNR (dB)  NB-IoT PDR(%)  LTE-M PDR(%)')
10 disp([SNR' NB_IoT' LTE_M'])
11 figure
12 % ----- Graph 1 -----
13 subplot(2,1,1)
14 plot(SNR,NB_IoT,'-o','LineWidth',2)
15 title('192512146 NB-IoT Packet Delivery Ratio vs SNR')
16 xlabel('Signal-to-Noise Ratio (dB)')
17 ylabel('Packet Delivery Ratio (%)')
18 grid on
19 % ----- Graph 2 -----
20
21 subplot(2,1,2)
22 plot(SNR,LTE_M,'-s','LineWidth',2,'color','g')
23 title('192512146 LTE-M Packet Delivery Ratio vs SNR')
24 xlabel('Signal-to-Noise Ratio (dB)')
25 ylabel('Packet Delivery Ratio (%)')
26 grid on
```



The Packet Delivery Ratio (PDR) of both NB-IoT and LTE-M increases as the Signal-to-Noise Ratio (SNR) improves. LTE-M achieves a slightly higher PDR than NB-IoT, indicating more reliable packet transmission under the same channel conditions.

Result:

1. The throughput performance of both NB-IoT and LTE-M was successfully simulated, and LTE-M achieved higher throughput than NB-IoT under the same network conditions.
2. The latency analysis showed that LTE-M provides lower communication delay compared to NB-IoT, making it more suitable for real-time IoT applications.
3. The Packet Delivery Ratio (PDR) of both technologies improved with increasing SNR, with LTE-M demonstrating slightly higher reliability than NB-IoT during data transmission.